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SECTION 01

INDUSTRY TALK REPORT

LECTURER: TS. DR. NOR AZIZAH ALI

Student Name	Matric No.
LAM YOKE YU	A23CS0233
TAN YI YA	A23CS0187
TEH RU QIAN	A23CS0191

1.0 Introduction

On 20 May 2026, Faiz Fablillah, a Senior Practitioner in Data & AI Governance, gave an industry talk. Topics covered in the event included data governance, data quality, analytics procedures and the expanding usage of artificial intelligence (AI) in businesses. It was divided into 4 parts, discussing the importance of data governance, explanation of the data management and governance concepts, highlighting how AI is transforming the industry and concluding with real-world examples. The purpose of the talk was to provide exposure to current industry practices involving data management and analytics. The importance of transforming raw data into meaningful insights that can support business operations, decision-making and automation. The industry talk was highly relevant to the topic of Business Intelligence (BI) and Analytics since it highlighted the relationship between data governance, data quality and the success of BI systems in modern organization.

2.0 Key Takeaway

One of the key points was the importance of data governance and quality in enterprise environments. Both are closely related to an Analytics Engineer's responsibilities. Mr Faiz highlighted that many BI problems come from upstream data issues like inconsistent sources, insufficient integration and poorly managed data pipelines, rather than dashboard construction itself. Analytics engineers are responsible for developing reliable data pipelines, processing raw data into analysis-ready datasets using tools such as SQL to ensure data integrity and accuracy. It was also pointed out that many organisations still rely on manual data cleaning, which increases inefficiency and mistake risk. The discussion also highlighted the importance of automation, governance frameworks, data ownership, and compliance guidelines in ensuring secure, accurate, and well-managed data for analytics and decision-making.

Another point is the importance of analytics in minimizing the gap between raw data and business decision-making. Data alone cannot add value unless it is managed, shared and evaluated across departments effectively. Mr Faiz highlighted that some organizations generate large amounts of data but fail to make it accessible for analytical purposes, limiting the effectiveness of Business Intelligence initiatives. The concept of master data established, which refers to critical organizational data shared across different departments while only maintaining the most latest and consistent records. Beside that, the talk discussed the differences between Generative AI and Agentic AI. Generative AI is primarily concerned with generating responses and reasoning, whereas Agentic AI may do tasks automatically. He addressed why many businesses are currently transitioning to tighter data governance procedures. In the early stages of digital transformation, organizations mainly focused on adopting BI tools and generating reports. However, as the amount of data and analytics used increased, the need for governance, data consistency and productive data management became more widely accepted across industries. The speaker explained the distinction between data governance and data management and emphasized how both functions complement each other within an organization. Data governance

is responsible for setting strategic direction, establishing policies and standards, assigning decision-making authority and measuring outcomes. In contrast, data management focuses on execution by implementing processes, operating systems, and delivering results based on the rules and objectives established through governance. The speaker also highlighted the importance of maintaining high data quality and introduced six key dimensions used to evaluate it, which are uniqueness, validity, timeliness, accuracy, completeness, and consistency. To support reliable and trustworthy data usage, organizations should maintain metadata (data that describes other data), monitor data lineage (tracking data from its source to its final form) and develop a data catalog as an organized inventory of all data assets across the organization. In addition, implementing data security and access control is essential to protect information assets and ensure regulatory compliance through practices such as Role-Based Access Control (RBAC), data classification and compliance management. Finally, the speaker introduced Master Data Management (MDM), commonly referred to as creating a “golden record,” which represents a single, trusted, and consistent version of critical organizational data.

The speaker also shared two real world example case studies to demonstrate how effective data governance and management practices can improve business intelligence outcomes. In the first case study involving Malaysian bank, which faced challenges caused by multiple disconnected systems containing conflicting customer information, resulting in inconsistent reporting and high operational costs for data reconciliation. To address these issues, Master Data Management (MDM) together with data quality rules and automated monitoring processes are implemented. By creating a centralized Golden Record and introducing real-time data validation and cleansing, the bank successfully established a single, consistent customer view across all channels, reduced regulatory risk by 40% and accelerated BI delivery to support faster and more reliable decision-making. The second case study on Telco company’s BI transformation, which has over 200 dashboards that users no longer trusted due to inconsistent reporting. The solution involved implementing data lineage, introducing data quality scorecards and developing a data catalog to improve transparency and confidence in data sources. As a result, the company reduced its dashboards to 50 trusted reports and achieved a threefold increase in user adoption and engagement.

3.0 Reflection and Critical Insights

The most interesting and valuable point from the talk was the notion of unused data in organisations. Data often remains unused for many reasons, including privacy concerns, ownership issues, or simply because people are unaware that the data exists. This highlighted the importance of data governance in persuading data owners that their data will be securely protected and properly managed. In addition, maintaining metadata or a data catalogue is essential so that all organisational data is acknowledged and accessible when needed. This is especially important in enterprise environments where large amounts of messy and unstructured data are generated and stored.

Another interesting issue raised by the speaker was how organisations can ensure that the outputs of data engineering processes, such as dashboards and reports, are accountable and trustworthy. It is important that results are traceable from the original data source, through every stage of the data engineering pipeline, until the final output. This traceability ensures that organisations can explain and justify the insights presented to decision-makers.

In business intelligence, dashboards and analytics are used to support business decisions, but their value depends heavily on the quality, governance, and reliability of the underlying data. Analytics engineering is crucial in the business ecosystem as it provides a bridge between data engineering and data analysis. According to the speaker, this is where data governance plays the most important role. As this is the last gatekeeper that the data goes through before it reaches out to the public, or in other terms the business world.

The speaker mentioned that data engineering and data governance are different and sometimes disagree with each other, as data engineers often prioritise faster pipeline development, while governance focuses on compliance and accountability. As a data engineering student, I believe it is important to integrate data governance into the development process instead of treating it as an afterthought. This would lead to more transparent and accountable analytical products.

This inspired us to further explore how organisations can balance fast data pipeline development with strong data governance practices. We should also learn more about how metadata management, data lineage and governance frameworks can be implemented in business intelligence systems.

4.0 Conclusion

In conclusion, the session has been an interesting and valuable talk that improved the practical understanding of what data governance related work is applied in real organisations. The session highlighted that successful business intelligence systems were very different from what we learned from attending lectures, and that communication skills were very important throughout the workflow. One of the most impactful lessons from the talk was the importance of balancing technical development with governance and accountability. Before attending the session, data engineering was viewed as building pipelines and processing data efficiently. However, the speaker showed that without proper governance, metadata management, data lineage, and security controls analytical outputs may become unreliable and difficult to trust.

The talk strengthened our interest in pursuing a career in data engineering, analytics engineering or business intelligence. It also encouraged us to expand my academic focus beyond technical skills such as SQL and analytics tools to include data governance, data quality management and AI governance. Overall, the talk motivated us to develop a more comprehensive understanding of the data industry.